

T0 Model: Unified Neutrino Formula Structure

Contents

.1 Preamble: Scientific Integrity 1

.2 Neutrinos as "Near-Massless Photons": The T0 Photon Analogy 2

 Photon-Neutrino Correspondence 2

 Double ξ -Suppression from Photon Analogy 3

.3 Neutrino Oscillations 4

 Geometric Phases as Oscillation Mechanism 5

.4 Fundamental Constants and Units 7

 Base Parameters 7

 Unit Conventions 7

.5 Charged Lepton Reference Masses 8

 Precise Experimental Values (PDG 2024) 8

.6 Neutrino Quantum Numbers (T0 Hypothesis) 8

 Postulated Quantum Number Assignment 8

 Geometric Factors 9

.7 Neutrino Mass Formula 9

 T0 Hypothesis: Equal Masses with Geometric Phases 9

.8 Plausible Target Value Based on Empirical Data 10

 Derivation from Measurements 10

.9 Experimental Comparison 11

 Current Experimental Upper Limits (2025) 11

 Safety Margins for T0 Hypothesis 12

.10 Consistency Checks and Validation 13

 Dimensional Analysis 13

 Mathematical Consistency 13

 Experimental Validation 14

Abstract

This document presents a mathematically consistent formula structure for neutrino calculations within the T0 model, based on the hypothesis of equal masses for all flavor states (ν_e, ν_μ, ν_τ). The neutrino mass is derived from the photon analogy ($\frac{\xi^2}{2}$ -suppression), and oscillations are explained by geometric phases based on $T_x \cdot m_x = 1$, with quantum numbers (n, ℓ, j) determining phase differences. A plausible target value for the neutrino mass ($m_\nu = 15$ meV) is derived from empirical data (cosmological constraints). The T0 model is based on speculative geometric harmonies without empirical support and is highly likely to be incomplete or incorrect. Scientific integrity requires a clear distinction between mathematical correctness and physical validity.

1 Preamble: Scientific Integrity

Warning

CRITICAL LIMITATION: The following formulas for neutrino masses are **speculative extrapolations** based on the untested hypothesis that neutrinos follow geometric harmonies and all flavor states have equal masses. This hypothesis has **no empirical basis** and is highly likely to be incomplete or incorrect. The mathematical formulas are nonetheless internally consistent and error-free.

Scientific Integrity Requires:

- Honesty about the speculative nature of predictions
- Mathematical correctness despite physical uncertainty
- Clear separation between hypotheses and verified facts

.2 Neutrinos as "Near-Massless Photons": The T0 Photon Analogy

Speculation

Fundamental T0 Insight: Neutrinos can be understood as "damped photons."

The remarkable similarity between photons and neutrinos suggests a deeper geometric kinship:

- **Speed:** Both propagate at nearly the speed of light
- **Penetration:** Both have extreme penetration capabilities
- **Mass:** Photon is exactly massless, neutrino is nearly massless
- **Interaction:** Photon interacts electromagnetically, neutrino interacts weakly

Photon-Neutrino Correspondence

Important

Physical Parallels:

$$\text{Photon: } E^2 = (pc)^2 + 0 \quad (\text{perfectly massless}) \quad (1)$$

$$\text{Neutrino: } E^2 = (pc)^2 + \left(\sqrt{\frac{\xi^2}{2}} mc^2 \right)^2 \quad (\text{nearly massless}) \quad (2)$$

Speed Comparison:

$$v_\gamma = c \quad (\text{exact}) \quad (3)$$

$$v_\nu = c \times \left(1 - \frac{\xi^2}{2} \right) \approx 0.9999999911 \times c \quad (4)$$

The speed difference is only 8.89×10^{-9} – practically unmeasurable!

Double ξ -Suppression from Photon Analogy

T0 Hypothesis: Neutrino = Photon with Geometric Double Damping

If neutrinos are "near-photons," two suppression factors arise:

- **First ξ Factor:** "Near massless" (like a photon, but not perfect)
- **Second ξ Factor:** "Weak interaction" (geometric coupling)
- **Result:** $m_\nu \propto \frac{\xi^2}{2}$, consistent with the speed difference $v_\nu = c \times \left(1 - \frac{\xi^2}{2}\right)$

Interaction Strength Comparison:

$$\sigma_\gamma \sim \alpha_{\text{EM}} \approx \frac{1}{137} \quad (5)$$

$$\sigma_\nu \sim \frac{\xi^2}{2} \times G_F \approx 8.888888 \times 10^{-9} \quad (6)$$

The ratio $\sigma_\nu/\sigma_\gamma \sim \frac{\xi^2}{2}$ confirms the geometric suppression!

.3 Neutrino Oscillations

Important

Neutrino Oscillations: Neutrinos can change their identity (flavor) during flight – a phenomenon known as neutrino oscillation. A neutrino produced as an electron neutrino (ν_e) can later be detected as a muon neutrino (ν_μ) or tau neutrino (ν_τ) and vice versa.

In standard physics, this behavior is described by the mixing of mass eigenstates (ν_1, ν_2, ν_3) connected to flavor states (ν_e, ν_μ, ν_τ) via the PMNS matrix (Pontecorvo-Maki-Nakagawa-Sakata):

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = U_{\text{PMNS}} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}, \quad (7)$$

where U_{PMNS} is the mixing matrix.

Oscillations depend on mass differences $\Delta m_{ij}^2 = m_i^2 - m_j^2$ and mixing angles. Current experimental data (2025) provide:

$$\Delta m_{21}^2 \approx 7.53 \times 10^{-5} \text{ eV}^2 \quad [\text{Solar}] \quad (8)$$

$$\Delta m_{32}^2 \approx 2.44 \times 10^{-3} \text{ eV}^2 \quad [\text{Atmospheric}] \quad (9)$$

$$m_\nu > 0.06 \text{ eV} \quad [\text{At least one neutrino, } 3\sigma] \quad (10)$$

Implications for T0:

- The T0 model postulates equal masses for flavor states $(\nu_e, \nu_\mu, \nu_\tau)$, implying $\Delta m_{ij}^2 = 0$, which is incompatible with standard oscillations.
- To explain oscillations, the T0 model uses geometric phases based on $T_x \cdot m_x = 1$, with quantum numbers (n, ℓ, j) determining phase differences.

Geometric Phases as Oscillation Mechanism

Speculation

T0 Hypothesis: Geometric Phases for Oscillations

To reconcile the hypothesis of equal masses ($m_{\nu_e} = m_{\nu_\mu} = m_{\nu_\tau} = m_\nu$) with neutrino oscillations, it is speculated that oscillations in the T0 model are caused by geometric phases rather than mass differences. This is based on the T0 relation:

$$T_x \cdot m_x = 1,$$

where $m_x = m_\nu = 4.54 \text{ meV}$ is the neutrino mass, and T_x is a characteristic time or frequency:

$$T_x = \frac{1}{m_\nu} = \frac{1}{4.54 \times 10^{-3} \text{ eV}} \approx 2.2026 \times 10^2 \text{ eV}^{-1} \approx 1.449 \times 10^{-13} \text{ s}.$$

The geometric phase is determined by the T0 quantum numbers (n, ℓ, j) :

$$\phi_{\text{geo},i} \propto f(n, \ell, j) \cdot \frac{L}{E} \cdot \frac{1}{T_x},$$

where $f(n, \ell, j) = \frac{n^6}{\ell^3}$ (or 1 for $\ell = 0$) are the geometric factors:

$$f_{\nu_e} = 1, \quad (11)$$

$$f_{\nu_\mu} = 64, \quad (12)$$

$$f_{\nu_\tau} = 91.125. \quad (13)$$

Calculated Phase Differences:

$$\phi_{\nu_e} \propto 1 \cdot \frac{L}{E} \cdot \frac{1}{T_x}, \quad (14)$$

$$\phi_{\nu_\mu} \propto 64 \cdot \frac{L}{E} \cdot \frac{1}{T_x}, \quad (15)$$

$$\phi_{\nu_\tau} \propto 91.125 \cdot \frac{L}{E} \cdot \frac{1}{T_x}. \quad (16)$$

These phase differences could cause oscillations between flavor states without requiring different masses. The exact form of the oscillation probability requires further development but remains highly speculative.

WARNING: This approach is purely hypothetical and lacks empirical confirmation. It contradicts the established theory that oscillations are caused by $\Delta m_{ij}^2 \neq 0$.

.4 Fundamental Constants and Units

Base Parameters

T0 Base Constants:

$$\xi = \frac{4}{3} \times 10^{-4} \approx 1.333333 \times 10^{-4} \quad [\text{dimensionless}] \quad (17)$$

$$\frac{\xi^2}{2} = \frac{\left(\frac{4}{3} \times 10^{-4}\right)^2}{2} \approx 8.888888 \times 10^{-9} \quad [\text{dimensionless}] \quad (18)$$

$$v = 246.22 \text{ GeV} \quad [\text{Higgs VEV}] \quad (19)$$

$$\hbar c = 0.19733 \text{ GeV}\cdot\text{fm} \quad [\text{Conversion constant}] \quad (20)$$

$$T_x = \frac{1}{4.54 \times 10^{-3} \text{ eV}} \approx 2.2026 \times 10^2 \text{ eV}^{-1} \approx 1.449 \times 10^{-13} \text{ s} \quad [\text{T0 Mass}] \quad (21)$$

Unit Conventions

Important

Consistent Unit Hierarchy:

$$\text{Standard: GeV} \quad (22)$$

$$\text{Submultiples: } 1 \text{ eV} = 10^{-9} \text{ GeV} \quad (23)$$

$$1 \text{ meV} = 10^{-12} \text{ GeV} = 10^{-3} \text{ eV} \quad (24)$$

$$\text{Masses: } m[\text{GeV}/c^2] = E[\text{GeV}]/c^2 \approx E[\text{GeV}] \text{ (natural units)} \quad (25)$$

$$\text{Time: } 1 \text{ eV}^{-1} \approx 6.582 \times 10^{-16} \text{ s} \quad (26)$$

.5 Charged Lepton Reference Masses

Precise Experimental Values (PDG 2024)

Experimental

Verified Particle Masses:

$$m_e = 0.51099895000 \times 10^{-3} \text{ GeV} = 510.99895 \text{ keV} \quad (27)$$

$$m_\mu = 105.6583745 \times 10^{-3} \text{ GeV} = 105.6583745 \text{ MeV} \quad (28)$$

$$m_\tau = 1776.86 \times 10^{-3} \text{ GeV} = 1.77686 \text{ GeV} \quad (29)$$

Unit Conversion to eV:

$$m_e = 510998.95 \text{ eV} = 510998950 \text{ meV} \quad (30)$$

$$m_\mu = 105658374.5 \text{ eV} \quad (31)$$

$$m_\tau = 1776860000 \text{ eV} \quad (32)$$

.6 Neutrino Quantum Numbers (T0 Hypothesis)

Postulated Quantum Number Assignment

Speculation

Hypothetical Neutrino Quantum Numbers:

$$\nu_e : \quad n = 1, \ell = 0, j = 1/2 \quad [\text{Ground state neutrino}] \quad (33)$$

$$\nu_\mu : \quad n = 2, \ell = 1, j = 1/2 \quad [\text{First excitation}] \quad (34)$$

$$\nu_\tau : \quad n = 3, \ell = 2, j = 1/2 \quad [\text{Second excitation}] \quad (35)$$

Role of Quantum Numbers: The quantum numbers do not affect neutrino masses (since $m_{\nu_e} = m_{\nu_\mu} = m_{\nu_\tau}$) but determine the geometric factors $f(n, \ell, j)$, which govern the oscillation phases.

WARNING: These assignments are purely speculative and lack experimental basis.

Geometric Factors

T0 Geometric Factors:

$$f(n, \ell, j) = \frac{n^6}{\ell^3} \quad \text{for } \ell > 0 \quad (36)$$

$$f(1, 0, j) = 1 \quad \text{for } \ell = 0 \text{ (special case)} \quad (37)$$

Calculated Values:

$$f_{\nu_e} = f(1, 0, 1/2) = 1 \quad (38)$$

$$f_{\nu_\mu} = f(2, 1, 1/2) = \frac{2^6}{1^3} = 64 \quad (39)$$

$$f_{\nu_\tau} = f(3, 2, 1/2) = \frac{3^6}{2^3} = \frac{729}{8} = 91.125 \quad (40)$$

.7 Neutrino Mass Formula

T0 Hypothesis: Equal Masses with Geometric Phases

Speculation

T0 Hypothesis: Equal Neutrino Masses with Geometric Phases

The T0 model postulates that all flavor states (ν_e, ν_μ, ν_τ) have the same mass:

$$m_{\nu_e} = m_{\nu_\mu} = m_{\nu_\tau} = m_\nu = 4.54 \text{ meV}.$$

The mass is derived from the photon analogy:

$$m_\nu = \frac{\xi^2}{2} \times m_e = (8.888888 \times 10^{-9}) \times (0.51099895 \times 10^{-3} \text{ GeV}) = 4.54 \text{ meV}.$$

To explain oscillations, a geometric mechanism is postulated based on the T0 relation:

$$T_x \cdot m_x = 1, \quad m_x = 4.54 \text{ meV}, \quad T_x \approx 2.2026 \times 10^2 \text{ eV}^{-1} \approx 1.449 \times 10^{-13} \text{ s}.$$

The oscillation phases are determined by geometric factors $f(n, \ell, j)$:

$$\phi_{\text{geo},i} \propto f_{\nu_i} \cdot \frac{L}{E} \cdot \frac{1}{T_x},$$

where $f_{\nu_e} = 1$, $f_{\nu_\mu} = 64$, $f_{\nu_\tau} = 91.125$.

Rationale:

- The mass 4.54 meV is consistent with the cosmological constraint ($\Sigma m_\nu = 0.01362 \text{ eV} < 0.07 \text{ eV}$).
- Geometric phases enable oscillations without mass differences, supporting the equal-mass hypothesis.
- This hypothesis is highly speculative and lacks empirical confirmation.

Formula: $m_{\nu_i} = 4.54 \text{ meV}$

Total Mass:

$$\Sigma m_\nu = 3 \times 4.54 \text{ meV} = 13.62 \text{ meV} = 0.01362 \text{ eV}$$

Comparison with Plausible Target Value:

- ν_e, ν_μ, ν_τ : 4.54 meV vs. 15 meV (Agreement: 30.3%)
- Σm_ν : 13.62 meV vs. 45 meV (Deviation: Factor ≈ 3.30)

Warning

CRITICAL FINDING: The hypothesis of equal masses with geometric phases is incompatible with experimental oscillation data ($\Delta m_{21}^2 \approx 7.53 \times 10^{-5} \text{ eV}^2$, $\Delta m_{32}^2 \approx 2.44 \times 10^{-3} \text{ eV}^2$), as it implies $\Delta m_{ij}^2 = 0$. The geometric approach is purely speculative and requires further theoretical and experimental validation.

.8 Plausible Target Value Based on Empirical Data

Derivation from Measurements

Experimental

Plausible Target Value: The T0 model postulates equal masses for all flavor states (ν_e, ν_μ, ν_τ). Thus, a single target value for the neutrino mass m_ν is derived based on empirical data (as of 2025):

- Cosmological Constraint: $\Sigma m_\nu = 3m_\nu < 0.07 \text{ eV} \implies m_\nu < 23.33 \text{ meV}$.
- Oscillation Data: $\Delta m_{21}^2 \approx 7.53 \times 10^{-5} \text{ eV}^2$, $\Delta m_{32}^2 \approx 2.44 \times 10^{-3} \text{ eV}^2$, typically requiring different masses. The T0 model bypasses this via geometric phases.
- Plausible Target Value: $m_\nu \approx 15 \text{ meV}$, lying between the solar (8.68 meV) and atmospheric scales (50.15 meV) and satisfying the cosmological constraint:

$$\Sigma m_\nu = 3 \times 15 \text{ meV} = 45 \text{ meV} = 0.045 \text{ eV} < 0.07 \text{ eV}.$$

Rationale:

- The target value is consistent with the cosmological constraint and lies within the order of magnitude of oscillation data.

- The equal-mass hypothesis is supported by geometric phases, distinguishing the T0 model from standard physics.
- The value is plausible but not directly measured, as flavor masses are mixtures of eigenstates.
- The T0 mass (4.54 meV) is below the target value (30.3%) but also cosmologically consistent.

.9 Experimental Comparison

Current Experimental Upper Limits (2025)

Experimental

Experimental Limits:

$$m_{\nu_e} < 0.45 \text{ eV} \quad [\text{KATRIN, 90\% CL}] \quad (41)$$

$$m_{\nu_\mu} < 0.17 \text{ MeV} \quad [\text{Muon decay, indirect}] \quad (42)$$

$$m_{\nu_\tau} < 18.2 \text{ MeV} \quad [\text{Tau decay, indirect}] \quad (43)$$

$$\Sigma m_\nu < 0.07 \text{ eV} \quad [\text{DESI+Planck, 95\% CL}] \quad (44)$$

$$\Delta m_{21}^2 \approx 7.53 \times 10^{-5} \text{ eV}^2 \quad [\text{Solar}] \quad (45)$$

$$\Delta m_{32}^2 \approx 2.44 \times 10^{-3} \text{ eV}^2 \quad [\text{Atmospheric}] \quad (46)$$

$$m_\nu > 0.06 \text{ eV} \quad [\text{At least one neutrino, } 3\sigma] \quad (47)$$

Safety Margins for T0 Hypothesis

Table 1: Safety Margins of the T0 Hypothesis Against Experimental Limits

Parameter	T0 Mass (4.54 meV)	Target Value (15 meV)
m_{ν_e} vs 0.45 eV	99200×	30×
m_{ν_μ} vs 0.17 MeV	3.74E7×	11333×
m_{ν_τ} vs 18.2 MeV	4.01E9×	1.21E6×

Parameter	T0 Mass (4.54 meV)	Target Value (15 meV)
Σm_ν vs 0.07 eV	$5.14\times$	$1.56\times$
Σm_ν vs 0.06 eV	$4.41\times$	$1.33\times$

Important

T0 Hypothesis:

- The T0 mass (4.54 meV) is consistent with cosmological constraints ($\Sigma m_\nu = 0.01362 \text{ eV} < 0.07 \text{ eV}$) and lies below the target value (15 meV, 30.3%).
- Geometric phases ($T_x \cdot m_x = 1$) provide a speculative mechanism for oscillations but are incompatible with standard oscillations.
- Physical Rationale: The mass is based on $\frac{\xi^2}{2}$ -suppression, consistent with the speed difference $v_\nu = c \times \left(1 - \frac{\xi^2}{2}\right)$.

.10 Consistency Checks and Validation

Dimensional Analysis

Dimensional Consistency:

$$[\xi] = 1 \quad \text{dimensionless} \quad (48)$$

$$[m_e] = \text{GeV} \quad \text{energy/mass} \quad (49)$$

$$\left[\frac{\xi^2}{2} \times m_e\right] = \text{GeV} \quad \text{energy/mass} \quad (50)$$

$$[f_{\nu_i}] = 1 \quad \text{dimensionless} \quad (51)$$

$$[m_\nu] = \text{eV} \quad (\text{fixed mass}) \quad (52)$$

$$[T_x] = \text{eV}^{-1} \quad (\text{time}) \quad (53)$$

All formulas are dimensionally consistent.

Mathematical Consistency

Important

Consistency of the Hypothesis:

- The formula $m_\nu = \frac{\xi^2}{2} \times m_e = 4.54 \text{ meV}$ is physically grounded in the photon analogy and consistent with the speed difference.
- Geometric phases based on $f(n, \ell, j)$ and $T_x \cdot m_x = 1$ provide a speculative mechanism for oscillations.
- No free parameters except ξ , simplifying the theory.

Experimental Validation

Experimental

Validation Status (as of 2025):

- The T0 mass (4.54 meV) satisfies cosmological constraints ($\Sigma m_\nu = 0.01362 \text{ eV} < 0.07 \text{ eV}$) and is close to the target value (15 meV, 30.3%).
- Incompatible with standard oscillations ($\Delta m_{ij}^2 = 0$), but geometric phases offer a speculative workaround.
- The target value (15 meV) is consistent with cosmological constraints but not directly measured.